Classes and Objects

What is a class?

A **class** is a blueprint or template used to create objects. It defines the data and functions that an object can have.

Main components of a class

Attributes (data): Variables that store information, such as name, age, and height.

Methods (functions): Functions defined inside a class that perform actions or work with the object's attributes.

Logical structure: Classes can contain conditional statements (if, else, elif) and loops (for, while) to control program execution.

What is an object?

An **object** is an instance of a class. It allows us to access the attributes and methods defined by that class.

```
class Student:
# Attributes
def __init__(self, name, age):
        self.name = name
        self.age = age
# Method
def display(self):
        print("Name:", self.name)
        print("Age:", self.age)
        # Creating an object
```

```
s1 = Student("Rahul", 20)
# Accessing attributes and method
print(s1.name)
print(s1.age) s1.display()
```

Output:

textRahul 20 Name: Rahul Age: 20